

#Registration & Transaction Flow (Shortened Version)

---Where VirtualAccount is same as Bank's ETCS and VehicleEntry is same as TollPlazaSystem

Step 1: Vehicle Registration

The vehicle owner registers online through the Bank's Electronic Toll Collection System (ETCS) by submitting required vehicle details (RFID tag or registration number). The Bank's ETCS securely shares this information with the Toll Plaza System via APIs.

Step 2: Vehicle Verification

The Toll Plaza System validates the vehicle information and checks eligibility for ETC. Once verified, it completes the registration and sends confirmation back to the Bank's ETCS.

Step 3: Virtual Account Creation

After verification, the Bank's ETCS creates a dedicated Virtual Account for toll recharges. The account can be funded via the bank's mobile app, online bill payment portal, or branch counters. Virtual Account details are shared with the Toll Plaza System, and once funded, the vehicle becomes eligible to use ETC lanes.

Step 4: Toll Fee Recharge

The vehicle owner tops up the Virtual Account through available banking channels. The Bank's ETCS validates the balance and confirms ETC eligibility. The owner can view the balance at any time and may request a refund of unused funds to any bank account in Bangladesh.

Step 5: Vehicle Entry into ETC Lane

When the vehicle enters an ETC lane, the Toll Plaza System detects it in real time, validates registration and account details, and sends a toll authorization request to the Bank's ETCS.

Step 6: Toll Fee Deduction

The Bank's ETCS checks the available balance and, upon success, authorizes the transaction. The toll fee is deducted from the Virtual Account within the defined transaction time (e.g., 5 seconds).

Step 7: Vehicle Pass Confirmation

After confirmation, the Toll Plaza System opens the gate to allow passage. The vehicle owner receives an SMS notification of the toll deduction.

Step 9: Balance Inquiry

The vehicle owner can check the remaining Virtual Account balance anytime through the bank's mobile or online banking platforms.

#Settlement Flow

Step 10: Bank Reconciliation

- The Bank's ETCS collects all successful transactions and confirms the amounts to the Bank's ETCS system continually.

Step 11: End-of-Day Settlement

- At the end of the day, the Bank's ETCS transfers the collected toll funds to the Toll Plaza Authority's bank account at sonali bank PLC and finally Generate a-Challan to Bangladesh A-Challan System.
- Also Report to ETC authority

#Dispute Revesal

Step 12: Toll Fee Reversal

If the vehicle does not pass after authorization, the Toll Plaza System requests a reversal. The Bank's ETCS validates and refunds the toll amount to the Virtual Account and confirms the reversal.

Step 13: Vehicle De-Registration

The vehicle owner submits a de-registration request through the Bank's ETCS with required vehicle details. The request is forwarded to the Toll Plaza System, which processes and confirms the de-registration.

Electronic Toll Collection (ETC) System Specification

1. System Participants

- **VehicleOwner:** The end-user who registers and manages their toll account.
 - **BankETCS:** The central banking and ledger system for ETC transactions.
 - **TollPlazaSystem:** The local hardware/software at the toll gate (RFID readers, barriers).
 - **SonaliBank:** The clearing bank for fund transfers.
 - **BangladeshAChallanSystem:** The government treasury system for revenue reporting.
 - **ETCAuthority:** The regulatory body receiving final daily reports.
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2. Process Flows

2.1 Vehicle Registration

1. **VehicleOwner** submits vehicle details (RFID or registration) to **BankETCS**.
2. **BankETCS** shares vehicle info via API with the **TollPlazaSystem**.
3. **TollPlazaSystem** validates and confirms the registration back to **BankETCS**.
4. **BankETCS** sends a "Registration confirmed" notification to the **VehicleOwner**.

2.2 Virtual Account Creation

1. **BankETCS** performs an internal process to "Create virtual account."
2. **BankETCS** shares virtual account details with the **TollPlazaSystem**.
3. **TollPlazaSystem** acknowledges the account info.

2.3 Toll Fee Recharge

1. **VehicleOwner** requests a "Top up virtual account" via **BankETCS**.
2. **BankETCS** confirms the new balance and verifies ETC eligibility to the **VehicleOwner**.

2.4 Vehicle Entry into ETC Lane

1. As a vehicle enters, the **TollPlazaSystem** sends a "toll authorization request" to **BankETCS**.
2. **BankETCS** authorizes the transaction and deducts the fee.
3. **BankETCS** signals the **TollPlazaSystem** to "Open gate and send SMS notification."

2.5 Balance Inquiry

1. **VehicleOwner** sends a "Request balance check" to **BankETCS**.
2. **BankETCS** provides the "current balance" response.

3. Settlement Flow (End of Day)

1. **BankETCS** performs an internal audit to "Collect successful transactions."
2. **BankETCS** initiates "Transfer toll funds" to **SonaliBank**.
3. **BankETCS** triggers the **BangladeshAChallanSystem** to "Generate A-Challan report."
4. **BankETCS** submits a "daily report" to the **ETCAuthority**.